What is physical AI?

TECHNOLOGY / PHYSICAL AI | AI perceives the real world, decides and drives machine actions

Physical AI is a technology field that connects AI with robots, vehicles, factory equipment and other machines to operate in the real world. Sensors gather information; AI decisions are passed to controllers and motors.

Perceive → decide → act

Perceive: cameras and sensors identify the surrounding situation.
Decide: AI models interpret the situation and determine the next action.
Act: controllers drive motors and machines.
The system perceives changes after acting and repeats judgment and control.
Safety, response speed and precision support use in real environments.

The range of related industries

Connections extend beyond robots and components to semiconductors for computing, communications, electricity and other infrastructure. The degree of connection varies by company and business.

Japanese companies and physical AI

Separate robots and components from industries supporting computing infrastructure

Field	Company / securities code	Business connection
Robots	Kawasaki Heavy Industries (7012)	Industrial robots and automation equipment: machinery motion.
Precision reducers	Harmonic Drive Systems (6324)	Precision reducers used in robot joints and elsewhere; support high-precision motion.
Precision reducers	Nabtesco (6268)	Precision reducers for industrial robots; support joint drives.
Power / infrastructure	Mitsubishi Heavy Industries (7011)	Power-generation equipment and gas turbines; indirect connection through electricity supply supporting AI computing.
Industrial / social infrastructure	Hitachi (6501)	AI, control, power and industrial systems; factory and infrastructure automation.
Semiconductor equipment	Tokyo Electron (8035)	Equipment for making semiconductors; part of the supply chain supporting AI computing infrastructure.
Advanced semiconductors	Rapidus	Aims to establish domestic production of advanced logic chips; connection on the computing side.
AI / communications infrastructure	GMO Internet Group (9449)	AI and internet infrastructure; surrounding infrastructure rather than robots themselves.

These are connections between each company’s business and technology fields; they do not indicate physical AI’s contribution to earnings.

Technologies and industries supporting physical AI

Look beyond finished products to supply chains for perception, decisions and control

Role	Main technology / equipment	Required performance
Brain	AI chips / AI models	Computing performance / energy efficiency
Computing / communications foundation	Data centers / power / communications	Stable supply / communications quality
Senses	Cameras / sensors / recognition	Accuracy of information acquisition / adaptation to the environment
Nerves	Controllers / software	Response speed / safety
Muscles / joints	Motors / reducers / actuators	Precision / durability / mass production
Body	Robots / vehicles / industrial machines	Real-world operation / task performance

Combine technologies from chips to machine motion

Practical deployment requires not only AI’s decision-making ability, but components, control systems and infrastructure for accurate and safe movement in the real world.